5. ESTIMATION OF CHOLESTEROL BY ZAK'S METHOD

Aim:

To determine the amount of cholesterol present in the given test solution by Zak's method

Principle:

In this method, cholesterol is treated with concentrated sulphuric acid. Cholesterol in the presence of sulphuric acid undergoes dehydration to form 3, 5-cholestadiene. This is inturn oxidised and sulphonated to form red colouredCholestapolyenesulphonic acid in the presence of Fe3+ ions. The intensity of red colour formed is proportional to the amount of cholesterol present in the test solution. The colour intensity is measured by using a green filter at 540nm.

Reagents:

1. **Stock ferric chloride reagent:** 840mg of was weighed and dissolved in 100 ml of glacial acetic acid.
2. **Ferric chloride diluting reagent:** 8.5ml of stock ferric chloride was diluted to 100ml with glacial acetic acid.
3. **Standard cholesterol solution:** 100mg pure dry cholesterol was dissolved in 100ml of acetic acid.
4. **Working standard solution:** 10ml of the stock is diluted to 100ml using acetic acid.

   Concentration = 100 µg / ml.

Procedure:

0.5-2.5ml of working standard cholesterol solution is pipetted out into five different test tubes and labelled as S1-S5 with the concentration of 50-250µg. 1.0ml of the test solution is pipetted out into a test tube and labelled as T_1. Final volume is made upto 5.0 ml using $FeCl_3$ diluting reagent in all the tubes. A blank is also prepared simultaneously comprising of 5.0ml of $FeCl_3$ diluting reagent. 3.0ml of conc. H_2SO_4 is added to all the test tubes. The contents in the test tubes are mixed well and kept at room temperature for 20 minutes. The intensity of red colour developed is measured at 540nm using a colorimeter.

A standard graph is drawn with optical density in Y- axis vs. concentration of cholesterol in X axis. The amount of cholesterol present in the given test solution is then calculated.

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